The central result of loop quantum gravity is indeed that space is not continuous, that it is not infinitely divisible but made up of grains or ‘atoms of space’. These are extremely minute: a billion billion times smaller than the smallest atomic nuclei. The theory describes these ‘atoms of space’ in mathematical form, and provides equations which determine their evolution. They are called ‘loops’, or rings, because they are linked to each other, forming a network of relations which weaves the texture of space, like the rings of a finely woven immense chain mail.

Where are these quanta of space? Nowhere. They are not in a space because they are themselves the space. Space is created by the linking of these individual quanta of gravity. Once again the world seems to be less about objects than about interactive relationships.

But it’s the second consequence of the theory that is the most extreme. Just as the idea of a continuous space that contains things disappears, so the idea of an elementary and primal ‘time’ flowing regardless of things also vanishes. The equations describing grains of space and matter no longer contain the variable ‘time’. This doesn’t mean that everything is stationary and unchanging. On the contrary, it means that change is ubiquitous – but elementary processes cannot be ordered in a common succession of ‘instants’. At the minute scale of the grains of space, the dance of nature does not take place to the rhythm of the baton of a single orchestral conductor, at a single tempo: each process dances independently of its neighbours, to its own rhythm. The passage of time is internal to the world, is born in the world itself in the relationship between quantum events that comprise the world and are themselves the source of time.

The world described by the theory is thus further distanced from the one with which we are familiar. There is no longer space which ‘contains’ the world, and there is no longer time ‘in which’ events occur. There are only elementary processes wherein quanta of space and matter continually interact with each other. The illusion of space and time which continues around us is a blurred vision of this swarming of elementary processes, just as a calm, clear Alpine lake consists in reality of a rapid dance of myriads of minuscule water molecules.

Viewed in extreme close-up through an ultrapowerful magnifying glass, the penultimate image in our fifth lesson should show the granular structure of space: